

Expectation-Maximization Algorithm

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1 Introduction

We discuss about **Expectation-Maximization algorithm** (EM algorithm) in detail. EM algorithm is used to find the MLE or MAP of a given likelihood function with a latent variable. We derive and explain EM algorithm only for discrete latent variables, but EM for continuous variables can be derived in the same way, exchanging summation with integration.

2 Main Idea

2.1 Intuition

Let $\mathbf{X}$ be a dataset in which the n -th row represents $\mathbf{x}_n^\top$, and let $\mathbf{Z}$ be latent variables in which the n -th row represents $\mathbf{z}_n^\top$. Representing $\boldsymbol{\theta}$ as the model parameter, the log likelihood function is given by $\log p(\mathbf{X}|\boldsymbol{\theta}) = \log \{\sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})\}$.

Maximizing the log likelihood is straightforward if all variables are observed. However, the latent variables are hardly known. The information we can get about latent variable is the posterior distribution $p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta})$. So, instead of maximizing the whole log likelihood, we maximize the expectation of log-likelihood under the posterior distribution of latent variables. Calculating the expectation is the **E step** of the EM algorithm. The subsequent step, the **M step** of the EM algorithm, then maximizes the expectation calculated at the E step. These pair of steps give rise to the revised estimate of parameters $\boldsymbol{\theta}$.

2.2 Algorithm

Algorithm 1: EM algorithm

Input : joint distribution $p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})$

initial parameters $\boldsymbol{\theta}^{(\text{old})}$

dataset $\mathbf{X}$

Output: Final parameters $\boldsymbol{\theta}$

```
1 repeat
    // E step
2    $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(\text{old})}) \leftarrow \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{(\text{old})}) \log p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})$ 
    // M step
3    $\boldsymbol{\theta}^{(\text{new})} \leftarrow \arg \max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(\text{old})})$ 
4    $\boldsymbol{\theta}^{(\text{old})} \leftarrow \boldsymbol{\theta}^{(\text{new})}$ 
5    $\mathcal{L} \leftarrow \log p(\mathbf{X}|\boldsymbol{\theta})$ 
6 until convergence
7 return  $\boldsymbol{\theta}^{(\text{new})}$ 
```

If we want to use the prior distribution $p(\boldsymbol{\theta})$ of parameter $\boldsymbol{\theta}$, we slightly modify the algorithm; we change M step so that it maximizes $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(\text{old})}) + \log p(\boldsymbol{\theta})$.

2.3 Evidence Lower Bound

EM algorithm can be seen at another perspective, called **Evidence lower bound** (ELBO). The ELBO plays important role in modern deep generative models and *variational inferences*.

2.3.1 Decomposition of Likelihood

Let $q(\mathbf{Z})$ be a distribution over the latent variables. Then, the following decomposition holds for log likelihoods;

Theorem 2.3.1 (Decomposition of log-likelihood Function)

Let $p(\mathbf{X}|\boldsymbol{\theta})$ be a likelihood of dataset $\mathbf{X}$. Let $q(\mathbf{Z})$ be a distribution over the latent variables. Then, the following holds;

$$\log p(\mathbf{X}|\boldsymbol{\theta}) = \mathcal{L}(q, \boldsymbol{\theta}) + \text{KL}(q||p)$$

where

$$\mathcal{L}(q, \boldsymbol{\theta}) = \sum_{\mathbf{Z}} q(\mathbf{Z}) \log \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})}{q(\mathbf{Z})} \right\} \text{ and } \text{KL}(q||p) = - \sum_{\mathbf{Z}} q(\mathbf{Z}) \log \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta})}{q(\mathbf{Z})} \right\}.$$

Notice that the KL-divergence is always non-negative, i.e., $\text{KL}(q||p) \geq 0$. Using this, we can derive the following useful result: $\mathcal{L}(q, \boldsymbol{\theta})$ provides the lower bound for log-likelihood such that $\mathcal{L}(q, \boldsymbol{\theta}) \leq \log p(\mathbf{X}|\boldsymbol{\theta})$.

2.3.2 EM Revisited

We can derive EM algorithm using Theorem 2.3.1. We first maximize the lower bound $\mathcal{L}(q, \boldsymbol{\theta}^{(\text{old})})$ about $q(\mathbf{Z})$, fixing the parameter as $\boldsymbol{\theta}^{(\text{old})}$. Since the parameter is fixed, the log-likelihood stays fixed and does not fluctuate as $q(\mathbf{Z})$ changes. So, to maximize $\mathcal{L}(q, \boldsymbol{\theta}^{(\text{old})})$, we minimize the KL-divergence, which can be accomplished when $q(\mathbf{Z}) = p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{(\text{old})})$. We now maximize lower bound about $\boldsymbol{\theta}$ with fixed $q(\mathbf{Z})$ this time. Since lower bound increases and the KL-divergence is non-negative since $q(\mathbf{Z})$ is fixed while $\boldsymbol{\theta}$ is modified, this will increase the log-likelihood. Substituting $q(\mathbf{Z}) = p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{(\text{old})})$ into the equation about lower bound of Theorem 2.3.1, we get

$$\begin{aligned} \mathcal{L}(q, \boldsymbol{\theta}) &= \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{(\text{old})}) \log p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta}) - \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{(\text{old})}) \log p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}^{(\text{old})}) \\ &= \mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(\text{old})}) + \text{const} \end{aligned}$$

where the constant is independent of $\boldsymbol{\theta}$. Then, the E step corresponds to maximizing $\mathcal{L}(q, \boldsymbol{\theta})$ about $q(\mathbf{Z})$ and the M step corresponds to maximizing $\mathcal{L}(q, \boldsymbol{\theta})$ about $\boldsymbol{\theta}$.

3 Extensions of EM Algorithm

3.1 Generalized EM Algorithm

When the model gets complex, the E step or M step, or both, can be intractable. So, instead of directly maximizing $\mathcal{L}(q, \boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$, we try to make the lower bound increase by modifying $\boldsymbol{\theta}$. For example, we can use gradient based algorithms in M step.

3.2 Sequential EM Algorithm

To be continued...